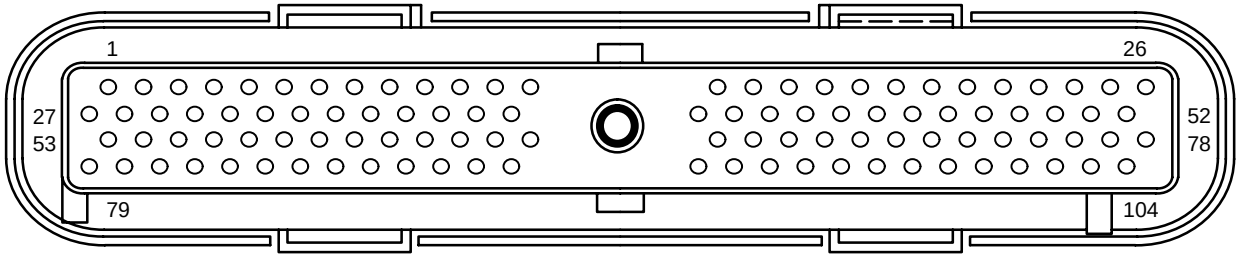
C175

A147
Powertrain Control Module (PCM) (12A650)



F104001

* 4.2L
** 4.6L
*** 5.4L
**** 5.4L Natural Gas Vehicle (NGV)
***** 6.8L
****** 7.3L

Pin	Circuit	Circuit function
1	1025 (OG/YE)	Coil On Plug (COP) 6 (12029), control
	*315 (VT/OG)	Shift solenoid B
	******–	not used
2	658 (PK/LG)	Malfunction Indicator Lamp (MIL), control
3	570 (BK/WH)	Ground
	*1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
	******3996 (RD/YE)	without memory, to, Manifold intake air heater relay
4	828 (VT/LB)	Voltage supplied at all times (overload protected)
	*–	not used
	**–	not used
	******284 (OG)	not used
5	–	not used
	******162 (LG/RD)	Parking brake switch (15A851)
6	237 (OG/YE)	Window lockout A
	*–	not used
7	–	not used
8	–	not used
	*1022 (DB/YE)	Intake Manifold Runner Control (IMRC) module, monitor
	*****1278 (WH/VT)	output, to, Glow plug relay

Pin	Circuit	Circuit function
9	29 (YE/WH)	Fuel pump/fuel gauge sender, Low fuel indicator
	*–	not used
	****–	not used
	****** 1279 (WH)	Fuel pump/fuel gauge sender
10	–	not used
	******308 (RD/OG)	Idle validation switch, control
11	315 (VT/OG)	Shift solenoid B
	*–	not used
12	911 (WH/LG)	Transmission Control Indicator Lamp (TCIL)
	*29 (YE/WH)	Fuel pump/fuel gauge sender
13	107 (VT)	Module programming signal
14	781 (OG/LB)	to 4L switch
	*–	not used
	**–	not used
	****–	not used
15	915 (PK/LB)	SCP Bus –
16	914 (TN/OG)	SCP Bus +
17	–	not used
	******1012 (OG/BK)	Digital Transmission Range (DTR) sensor (7F293), Digital Transmission Range (DTR) sensor
18	–	not used
19	–	not used
	******648 (WH/PK)	signal

Pin	Circuit	Circuit function
20	924 (BN/OG)	Powertrain Control Module (PCM) (12A650), Coasting clutch solenoid
	* –	not used
21	349 (DB)	Crankshaft position sensor (6C315) –
	******795 (DG)	Camshaft position sensor (6B288), to, Powertrain Control Module (PCM) (12A650)
22	350 (GY)	Crankshaft position sensor (6C315) +
	******–	not used
23	–	not used
	*****570 (BK/WH)	Ground
24	570 (BK/WH)	Ground
	******837 (YE/BK)	Accelerator pedal position sensor, to, Powertrain Control Module (PCM) (12A650)
25	875 (BK/LB)	Ground
26	1024 (LG/WH)	Coil On Plug (COP) 1 (12029), control
	*526 (PB/LG)	PCM, to, Ignition coil (12029)
	****** –	not used
27	1021 (LG/YE)	Coil On Plug (COP) 5 (12029), control
	*****1142 (WH/RD)	Coil On Plug (COP) 10, control
	237 (OG/YE)	Transmission Control Indicator Lamp (TCIL)
	****** –	not used
28	–	not used
	*****911 (WH/LG)	Transmission Control Indicator Lamp (TCIL)
	******643 (RD)	indicator, Water–in–fuel sensor
29	224 (TN/WH)	PCM, to transmission overdrive cancel switch
30	–	not used
	******553 (VT/LB)	Exhaust Back Pressure (EBP) sensor, to, Powertrain Control Module (PCM) (12A650)
31	–	not used
	******810 (RD/LG)	Brake pedal position switch (13480), input

Pin	Circuit	Circuit function
32	311 (DG/PK)	Knock sensor (12A699) –
	**** –	not used
	****** –	not used
33	–	not used
	*****676 (PK/OG)	Vehicle Speed Sensor (VSS) –
	******676 (PK/OG)	Vehicle Speed Sensor (VSS) –
34	1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
	* –	not used
	******1277 (WH/LG)	not used
35	392 (RD/LG)	Heated Oxygen Sensor (HO2S) #12 (9G444), input
	**** –	not used
	******1183 (WH/YE)	Generator, Power
36	968 (TN/LB)	Mass Air Flow (MAF) sensor (12B579), signal, return
	******1280 (GY/RD)	Fuel heater, to, Powertrain Control Module (PCM) (12A650)
37	923 (OG/BK)	Transmission Fluid Temperature (TFT) sensor, input
38	–	not used
	******354 (LG/RD)	Engine Oil Temperature (EOT) sensor, input
39	743 (GY)	Intake Air Temperature (IAT) sensor (12A697), input
40	238 (PG/YE)	Fuel pump, monitor
	******787 (PK/BK)	Fuel pump, monitor
41	347 (BK/YE)	A/C clutch relay, switched output
	**439 (TN/LG)	A/C pressure switch, signal
42	–	not used
	*1023 (DB/LG)	Intake Manifold Runner Control (IMRC) module, to, Powertrain Control Module (PCM) (12A650)
	******318 (GY/RD)	Exhaust Pressure Regulator (EPR), to, Powertrain Control Module (PCM) (12A650)

Pin	Circuit	Circuit function
43	–	not used
44	–	not used
45	1105 (YE/LB)	Cylinder head temperature sensor, to, Temperature, indicator
	****** –	not used
46	–	not used
	**367 (BN)	Intake Manifold Runner Control (IMRC) module
	*****679 (GY/BK)	Vehicle Speed Sensor (VSS) +
47	360 (BN/PK)	EGR Vacuum Regulator (EVR), control
	**** –	not used
	****** –	not used
48	648 (WH/PK)	Tachometer, signal
	******818 (GY/WH)	Injector Driver Module (IDM) power relay, to, Powertrain Control Module (PCM) (12A650)
49	1145 (LB/BK)	Digital Transmission Range (DTR) sensor, TR2
	******146 (WH/PK)	Digital Transmission Range (DTR) sensor, TR2
50	1146 (LG/RD)	Digital Transmission Range (DTR) sensor, to, Powertrain Control Module (PCM) (12A650)
	******145 (GY/BK)	Shift solenoid
51	570 (BK/WH)	Ground
52	1028 (WH/PK)	Coil On Plug (COP) 3 (12029), control
	*****1021 (LG/YE)	Coil On Plug (COP) 5 (12029), control
	*527 (RD/LB)	PCM, to, Ignition coil (12029)
	****** –	not used
53	1030 (DG/VT)	Coil On Plug (COP) 4 (12029), control
	*****1027 (PK/LB)	Coil On Plug (COP) 7 (12029), control
	* –	not used
	****** –	not used

Pin	Circuit	Circuit function
54	480 (VT/YE)	Torque Converter Clutch (TCC) solenoid
55	37 (YE)	Voltage supplied at all times (overload protected)
56	191 (LG/BK)	Powertrain Control Module (PCM) (12A650), EVAP canister purge valve
	*** –	not used
	****** –	not used
57	310 (YE/RD)	Knock sensor (12A699) –
	**** –	not used
	****** –	not used
58	–	not used
	******679 (GY/BK)	Vehicle Speed Sensor (VSS) +
59	970 (DG/WH)	Turbine Shaft Speed (TSS) sensor
	* & ** –	not used
	****** –	not used
60	74 (GY/LB)	Heated Oxygen Sensor (HO2S) #11 (9F472), input
	******1185 (YE)	Generator, Power
61	–	not used
	* & ** 393 (VT/LG)	Heated Oxygen Sensor (HO2S) #22 (9G444), input
	******151 (LB/BK)	Speed control switch, input
62	791 (RD/PK)	Fuel tank pressure transducer sensor (9C968)
	****1164 (LB)	Engine controls, Gas temperature sensor (CNG), Temperature
	****** –	not used
63	–	not used
	****141 (RD/PK)	Injector pressure sensor, except 1,9 TDI unit injector engine
64	199 (LB/YE)	Clutch pedal position switch, to, Powertrain Control Module (PCM) (12A650)
65	352 (BN/LG)	Differential Pressure Feedback EGR (DPFE) sensor, input
	**** –	not used
	******796 (LB)	Camshaft position sensor (6B288), signal, return

Pin	Circuit	Circuit function
66	1102 (YE/LG)	Cylinder–head temperature sensor (6G004), to, Powertrain Control Module (PCM) (12A650)
	******439 (TN/LG)	Modified vehicle power, to, Powertrain Control Module (PCM) (12A650)
67	91 (VT/WH)	Canister vent
	**** –	not used
	91 (VT/WH)	Power, Battery Power
68	679 (GY/BK)	Vehicle Speed Sensor (VSS) +
	*****1100 (GY/BK)	Fuel injector 9 (9F593), control
	****** –	not used
69	–	not used
	**883 (BK/LB)	Voltage supplied at all times (overload protected)
	331 (PK/YE)	A/C clutch relay, control
70	–	not used
	******464 (BK/PK)	Low coolant indicator, control
71	361 (RD)	Voltage supplied in Start and Run (overload protected)
72	561 (TN/RD)	Fuel injector 7, control
	* –	not used
	****** –	not used
73	559 (TN/BK)	Fuel injector 5, control
	****** –	not used
74	557 (BN/YE)	Fuel injector 3, control
	****** –	not used
75	555 (TN)	Fuel injector 1, control
	****** –	not used
76	570 (BK/WH)	Ground
	**** –	not used
77	570 (BK/WH)	Ground
78	1027 (PK/LB)	Coil On Plug (COP) 7 (12029), control
	*528 (PK/WH)	PCM, to, Ignition coil (12029)
	*****1026 (PK/WH)	Coil On Plug (COP) 2 (12029), control
	****** –	not used

Pin	Circuit	Circuit function
79	1029 (WH/LG)	Coil On Plug (COP) 8 (12029), control
	*911 (WH/LG)	Transmission Control Indicator Lamp (TCIL)
	******358 (LG/BK)	Manifold Absolute Pressure (MAP) sensor, to, Powertrain Control Module (PCM) (12A650)
80	926 (LB/OG)	Fuel pump relay, control
	******814 (WH/BK)	Injector Driver Module (IDM) power relay, control
81	925 (WH/YE)	to Electronic Pressure Control (EPC) solenoid
82	–	not used
	*****1141 (YE/BK)	Coil On Plug (COP) 4 (12029), control
83	264 (WH/LB)	Idle Air Control (IAC) valve (9F715), control
	******552 (YE/RD)	Injection Pressure Regulator (IPR), to, Powertrain Control Module (PCM) (12A650)
84	136 (DB/YE)	Powertrain Control Module (PCM) (12A650) – Output Shaft Speed (OSS) sensor (7M101)
	******970 (DG/WH)	Turbine Shaft Speed (TSS) sensor
85	795 (DG)	Camshaft position sensor (6B288), to, Powertrain Control Module (PCM) (12A650)
	****** –	not used
86	–	not used
87	94 (RD/BK)	Heated Oxygen Sensor (HO2S) #21 (9F472), input
	812 (DB/LG)	Injection Control Pressure (ICP) sensor
88	967 (LB/RD)	Mass Air Flow (MAF) sensor (12B579), output
	****** –	not used
89	355 (GY/WH)	Throttle Position Sensor (TPS) (9B989), input
90	351 (BN/WH)	Reference, voltage
91	359 (GY/RD)	signal, return
92	511 (LG)	Brake pedal position switch (13480), to stoplamps
93	387 (RD/WH)	Heated Oxygen Sensor (HO2S) #11 (9F472), heater
	****** –	not used
94	388 (YE/LB)	Heated Oxygen Sensor (HO2S) #21 (9F472), heater
	******926 (LB/OG)	Fuel pump relay, control

Pin	Circuit	Circuit function
95	389 (WH/BK)	Heated Oxygen Sensor (HO2S) #12 (9G444), heater
	**** –	not used
	******821 (BN/OG)	fuel delivery command, signal, output
96	–	not used
	* & **390 (TN/YE)	Heated Oxygen Sensor (HO2S) #22 (9G444), heater
	******817 (YE/LB)	cylinder identification, signal, output
97	361 (RD)	Voltage supplied in Start and Run (overload protected)
	* –	not used
98	562 (LB)	Fuel injector 8, control
	* –	not used
	******462 (PK)	Manifold intake air heater relay, control
99	560 (LG/OG)	Fuel injector 6, control
	****** –	not used

Pin	Circuit	Circuit function
100	558 (BN/LB)	Fuel injector 4, control
	******29 (YE/WH)	Fuel pump/fuel gauge sender
101	556 (WH)	Fuel injector 2, control
	******1086 (VT/OG)	Glow plug relay, to, Powertrain Control Module (PCM) (12A650)
102	–	not used
	1030 (DG/VT)	Coil On Plug (COP) 4 (12029), control
103	570 (BK/WH)	Ground
104	1026 (PK/WH)	Coil On Plug (COP) 2 (12029), control
	*****1028 (WH/PK)	Coil On Plug (COP) 3 (12029), control
	* –	not used
	****** –	not used